

LINGUERE, Senegal

WMO: 616270

Lat: 15.383N

Lon: 15.117W

Elev: 21

StdP: 101.07

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.7	17.0	-5.3	2.4	27.7	-3.6	2.8	28.1	6.4	26.7	5.6	26.9	1.5	40	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	16.5	43.3	22.6	42.1	22.5	41.0	22.5	28.1	34.7	27.6	33.8	27.2	33.1	2.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.3	30.7	26.2	21.6	30.2	25.9	21.2	30.0	90.1	35.3	88.0	33.9	86.1	33.1	32.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
5.4	4.5	4.0	12.5	45.7	1.2	1.1	11.6	46.5	10.8	47.2	10.1	47.8	9.2	48.6		
			6.9	30.0	1.6	1.4	5.8	30.9	4.8	31.7	3.9	32.5	2.8	33.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	29.8	25.3	27.3	30.0	31.5	32.7	32.6	30.9	29.6	29.5	31.4	29.7	26.5	(6)
(7)		DBStd	3.07	2.36	2.54	2.69	2.63	2.20	1.78	1.68	1.56	1.52	1.55	1.71	2.27	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	7209	473	486	621	644	705	677	647	607	585	663	592	511	(10)
(11)		CDD18.3	4168	214	252	362	394	447	427	389	349	335	405	342	252	(11)
(12)		CDH23.3	56497	2348	3140	5140	5932	6826	6282	5317	4388	4122	5626	4547	2828	(12)
(13)		CDH26.7	33811	1201	1792	3266	3999	4675	4061	2983	2129	1991	3397	2781	1536	(13)
(14)	Wind	WSAvg	2.0	2.1	2.2	2.1	2.2	2.2	2.3	2.2	1.7	1.5	1.5	1.6	2.0	(14)
(15)	Precipitation	PrecAvg	453	1	1	0	0	1	24	92	147	119	35	1	2	(15)
(16)		PrecMax	783	35	32	16	6	15	133	277	389	363	213	19	31	(16)
(17)		PrecMin	256	0	0	0	0	0	0	9	41	13	0	0	0	(17)
(18)		PrecStd	127	5	5	2	1	2	24	59	67	68	43	4	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.6	40.6	42.9	44.5	44.9	44.1	40.1	38.0	37.8	41.5	40.7	39.0	(19)
(20)			MCWB	18.3	19.0	21.4	23.3	23.0	23.7	25.0	25.5	26.9	23.6	22.7	20.4	(20)
(21)		2%	DB	35.7	38.5	41.4	42.9	43.4	42.1	38.5	36.2	35.9	40.2	39.5	37.1	(21)
(22)			MCWB	17.4	18.7	20.8	22.1	22.6	23.9	25.0	26.2	27.1	24.0	21.6	19.6	(22)
(23)		5%	DB	34.2	36.8	40.0	41.5	42.1	40.8	37.2	34.9	34.9	39.2	38.5	35.7	(23)
(24)			MCWB	17.0	18.3	20.7	21.7	22.8	24.1	25.2	26.2	27.1	23.8	21.1	19.1	(24)
(25)		10%	DB	32.7	35.2	38.4	40.0	40.7	39.3	36.0	33.8	33.9	37.9	37.3	34.1	(25)
(26)			MCWB	16.7	17.9	20.0	21.2	22.7	24.2	25.2	26.4	27.0	23.9	20.6	18.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	21.6	25.4	27.8	27.4	28.2	28.1	28.2	28.7	28.7	25.6	24.5	(27)
(28)			MCDWB	31.6	35.5	39.0	40.4	39.2	37.7	35.2	33.6	34.2	37.4	37.5	34.1	(28)
(29)		2%	WB	19.5	20.4	23.6	25.7	25.7	26.9	27.1	27.6	28.1	27.7	24.3	22.4	(29)
(30)			MCDWB	31.6	34.5	37.7	37.8	38.8	36.3	34.0	32.9	33.5	35.1	35.2	33.1	(30)
(31)		5%	WB	18.4	19.6	22.1	24.1	24.7	26.0	26.5	27.2	27.7	27.0	23.2	20.9	(31)
(32)			MCDWB	31.1	33.6	36.0	37.2	37.6	35.6	33.3	32.3	32.9	33.7	34.0	31.8	(32)
(33)		10%	WB	17.4	18.8	20.8	22.8	24.0	25.3	26.1	26.8	27.3	26.4	22.2	19.8	(33)
(34)			MCDWB	30.6	32.6	35.0	35.9	36.5	34.9	32.7	31.8	32.3	32.6	33.9	30.4	(34)
(35)	Mean Daily Temperature Range	MDBR	14.8	15.1	15.8	16.7	16.5	14.3	10.3	8.5	9.0	13.4	15.8	15.0	(35)	
(36)		5% DB	MCDBR	17.0	16.8	17.6	17.8	17.4	16.2	12.2	10.1	10.3	15.4	17.3	17.6	(36)
(37)			MCWBR	6.8	6.1	5.9	5.3	4.7	3.8	2.9	3.0	3.3	4.3	6.0	7.0	(37)
(38)		5% WB	MCDBR	15.4	15.2	17.0	16.9	16.7	13.6	10.0	8.8	9.6	12.6	15.8	15.0	(38)
(39)			MCWBR	6.6	6.0	6.8	6.6	5.3	4.5	3.3	3.3	3.6	3.9	6.7	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.532	0.565	0.642	0.677	0.729	0.771	0.725	0.648	0.613	0.606	0.516	0.504	(40)	
(41)		taud	1.797	1.728	1.585	1.568	1.478	1.476	1.587	1.747	1.812	1.796	1.916	1.904	(41)	
(42)		Ebn at Noon	755	754	711	687	642	608	640	699	723	713	766	768	(42)	
(43)		Edh at Noon	209	234	277	282	304	301	270	233	218	216	184	184	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.17	5.83	6.66	7.01	6.92	6.54	6.11	5.88	5.79	5.73	5.22	4.91	(44)	
(45)		RadStd	0.29	0.30	0.24	0.23	0.24	0.16	0.16	0.18	0.22	0.21	0.25	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page